

HW#7 KEY – (30 PTS)**REF: READ THEORY MANUAL SECTIONS §2.3-§2.6
MODULE 7 – LECTURE SLIDES****(10 PTS) 1: Reactor Theory Manual - Chapter 2 (page 2-31 §2.14), DO QUESTION #1**

Hint: For #1 – reference §2.5 ‘Exponential Power Equations’. Use the InHour curve or ‘One-Group’ approximation to relate ρ to τ . Please show all work!

(3 PTS)

Eq. 2.5: $P_f = P_o e^{t/\tau}$ so $\ln\left(\frac{P_f}{P_o}\right) = \frac{t}{\tau}$ where $t = 120 \text{ sec (2 min)}$

So: $\ln\left(\frac{1500}{100}\right) = \frac{120 \text{ sec}}{\tau}$ therefore reactor period $\tau = 44.3 \text{ sec}$

From Eq. 2.7, $\tau = 1.44 \times dt$; so doubling time (dt) = $\tau/1.44 = 30.8 \text{ sec}$

(2 PTS)

From Eq. 2.7. SUR (in decades per minute) = $26.1/\tau$ (in sec), so SUR = $26.1/44.3 = 0.589 \text{ DPM}$

(2 PTS)

Where Decade Time (DT)(min) = $1/\text{SUR} = 1/0.589 = 1.7 \text{ minutes per decade}$

(3 PTS)

Reactivity Addition calculation using One-Group In-hour Approximation:

From Eq. 2.10: $\rho = \frac{\beta_{eff}}{1 + \lambda_{eff}\tau}$ where $\beta_{eff} = 0.0073 \Delta K/K = 730 \text{ pcm}$ and $\lambda_{eff} = 0.1 \text{ sec}^{-1}$ for slightly supercritical.

So: if $\tau = 44.3 \text{ sec}$, then $\rho = 134 \text{ pcm} = 0.00134 \Delta K/K$

2. (3 PTS) Define Reactor Period ‘ τ ’.

The time it takes for reactor power to increase or decrease by a factor of e ($= 2.718$).

3. Given the “One-Group Approximation” to the InHour Equation (Eq. #2.10), if $\lambda_{eff} \sim 0.1 \text{ sec}^{-1}$ (approximation for “slightly critical”) & $\beta_{eff} = 730 \text{ pcm}$:

a) (4 PTS) What is the period τ (sec) if $\rho_{net} = 0.0030 \Delta k/k$ (300 pcm)?

*From Eq. 2.10: $\rho = \frac{\beta_{eff}}{1 + \lambda_{eff}\tau}$; and solving for τ we obtain a period of **14.3 seconds**.*

- b) **(3 PTS)** What is the Startup Rate (SUR) in Decades-per-minute (DPM)? How long will it take for reactor power to increase by a factor of ten (10)?

SUR (in decades per minute) = $26.1 / \tau$ (in sec), so $SUR = 26.1 / 14.3 = 1.83$ DPM

A factor of ten is one decade (x10), where Decade Time (DT) = $1 / SUR = 1 / 1.83 = 0.548$ min.

4. From the InHour curve:

- a) **(3 PTS)** What is the period τ (sec) if $\rho_{net} = 0.0030 \Delta k/k$?

$\rho_{net} = 0.0030 \Delta k/k = 300$ pcm. From InHour curve, $\tau \approx 9.3$ seconds.

- b) **(4 PTS)** Using this period, how long would it take to increase reactor power from 1 W to 1 kW?

Eq. 2.5: $P_f = P_o e^{t/\tau}$ so $\ln\left(\frac{P_f}{P_o}\right) = \frac{t}{\tau}$ where $\tau = 9.3$ sec

So: $\ln\left(\frac{1000}{1}\right) = \frac{t}{9.3}$ therefore time to increase $t = 64.2$ seconds.

5. **(3 PTS)** What is the difference (in seconds) for how long (t) it would take to increase power from 1 Watt to 1 kW using the two different periods from Problems #3a & #4a above?

Check their work. If they got either 3a or 4a incorrect, but correctly used their answer in this question, they receive the points for this question. However, a note should be provided referencing that their initial answer for 3a or 4a was incorrect.

From 3(a): $\tau = 14.3$ seconds, so from Eq. 2.5, $t = 98.8$ seconds

So, $\Delta t = 98.8 - 64.2 = 34.6$ second difference! Note – One Group Approximation not valid for larger reactivity values e.g. $\rho_{net} = 0.0030 \Delta k/k$!